

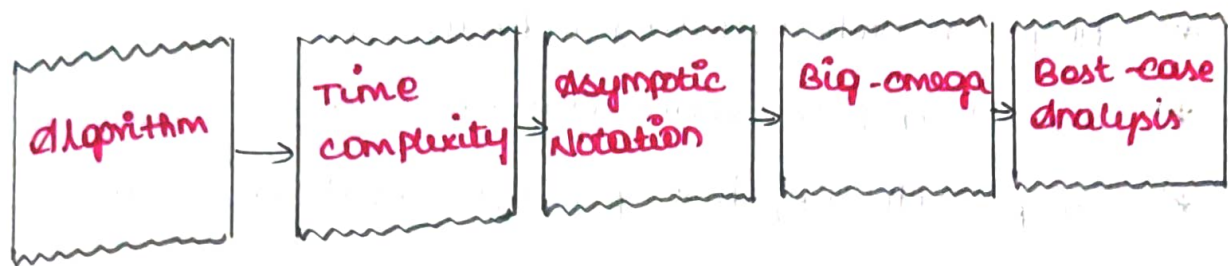
CSA0612 - Design and Analysis of Algorithm

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Ass no : 2 Concept Map

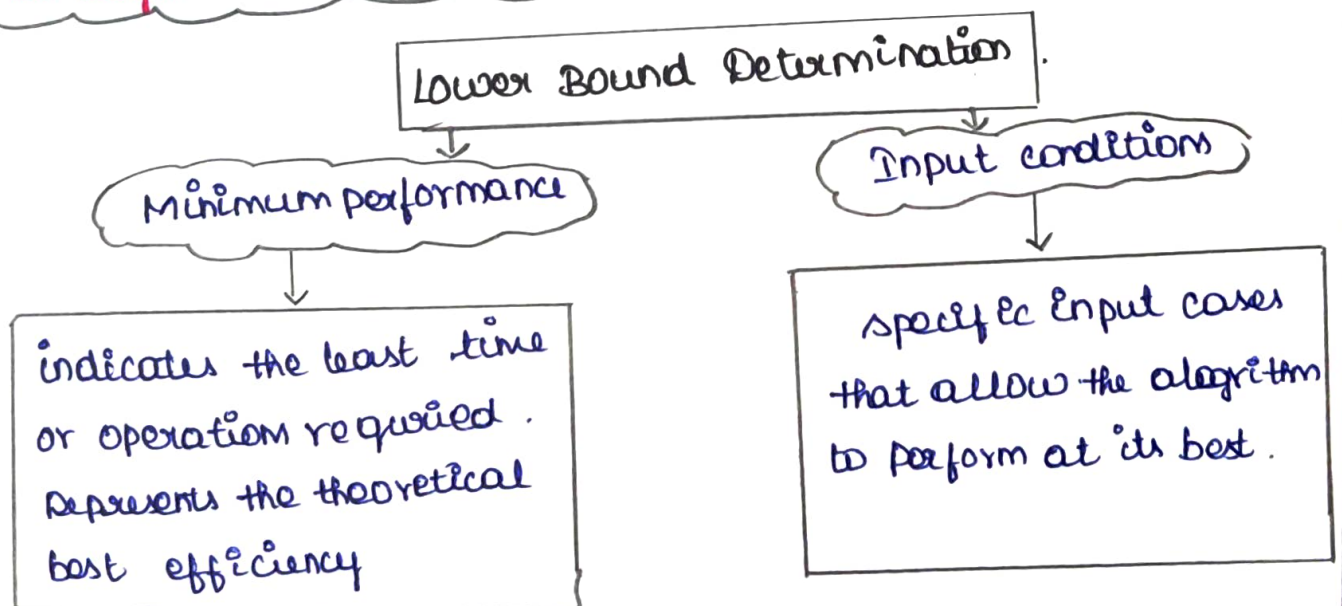
Q 53. Lower Bound and Best-case Analysis



How lower bound is determined?

- *1) Analyze the algorithm under the best possible input conditions
- *2) count the minimum number of operations executed
- *3) Express this minimum using Asymptotic Notation (Big-omega Ω)
- *4) This gives the lower bound of the time complexity.

Lower bound represents the minimum number of operations an algorithm must perform in the best possible scenario.



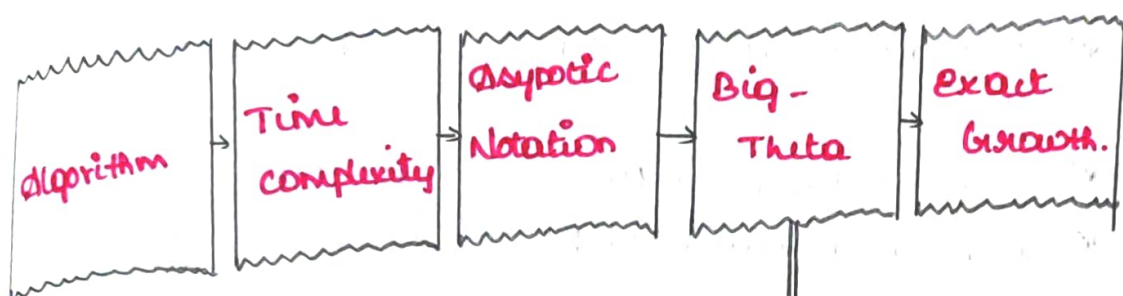
Importance of Best case Analysis :

- * Provides optimistic scenario of algorithm's performance.
- * Helps in understanding potential efficiency.
- * Useful in performance guarantees and comparison.
- * Helps identify best possible input arrangements.

Contributions to complete understanding :

- * Complete average and worst case Analysis.
- * Provides a performance range (best to worst)
- * Helps in algorithm's selection for favorable scenarios
- * Ensures a complete behavioral perspective of the algorithm.

Q54. Exact Growth and Tight Bound complexity



upper Bound

represent the maximum Number of operations. Algorithm Cannot gives faster than this

lower Bound.

Represents the maximum Number of operation algorithm cannot grow slower than this.

Big- θ (θ) = Tight Bound

$\theta(g(n)) = \{f(n) \mid c_1 g(n) \leq f(n) \leq c_2 g(n), \text{ for } n \geq n_0\}$ exactly describes the growth rate of the algorithm.

How Big- θ represents Tight Bound complexity?

- * Big- θ combines both upper bound (O) and lower bound (Ω).
- * It gives a tight bound where the algorithm grows at the same rate.
- * It excludes constant factors and lower order terms.

Why exact bound is Important?

- *) Gives the most precise analysis of algorithm
- *) eliminates ambiguity between best and worst bounds.
- *) crucial for performance-critical applications.
- *) Helps in accurate comparison between algorithms.

How to improve map performance?

- *) provides complete growth behaviour (both upper & lower bound).
- *) Helps in predicting performance accurately.
- *) aids in choosing the best algorithm for problem.
- *) essential for optimizing and scaling solutions.